

AI-Powered Influencer–Brand Matching via Multimodal Semantic Embeddings and Multi-Head Attention Profiling

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Abstract—Influencer selection in digital marketing remains predominantly metric-driven, relying on follower counts and engagement rates rather than genuine content–campaign alignment. This paper presents an AI-powered influencer–brand matching platform that addresses this gap by encoding each influencer’s complete post history into a 256-dimensional identity vector using a custom Multi-Head Attention Profiler trained on OpenCLIP (ViT-B/32) embeddings. The system was developed on a dataset of 33,935 Instagram influencers and 9.5 million posts across 8 content categories. The best model achieves 85.49% classification accuracy, 80.84% Precision@10, a Hit Rate@10 of 95.89%, and a mean reciprocal rank (MRR) of 0.8612. An embedding space separation ratio of 3.61x demonstrates a well-structured retrieval space. End-to-end pipeline validation on a real-world account confirms autonomous influencer onboarding. The platform integrates a React.js frontend, FastAPI backend, and Supabase PostgreSQL database with pgvector for real-time cosine similarity retrieval.

Index Terms—influencer marketing, multimodal embeddings, OpenCLIP, attention mechanism, recommender system, pgvector

I. INTRODUCTION

Influencer marketing has grown into one of the most impactful digital advertising strategies, particularly in emerging markets such as Pakistan, where social media adoption is accelerating but structured discovery infrastructure remains absent. Brands currently select collaborators based on follower counts, visible engagement metrics, or personal networks—signals that do not capture whether an influencer’s content identity genuinely aligns with a campaign’s positioning [6], [7].

Three structural problems limit current practice. First, selection is metric-driven rather than content-driven: quantitative signals cannot distinguish a fashion influencer who recently posted gym content from a genuine fitness creator. Second, capable micro-influencers (10,000–50,000 followers), who often achieve higher engagement rates and stronger niche authority, are systematically overlooked. Third, the broader collaboration workflow—booking, contract management, payment, and performance tracking—is handled through direct messages and informal bank transfers, creating trust gaps and disputes.

Recent advances in contrastive vision-language models, specifically OpenCLIP, enable encoding both images and captions into a shared high-dimensional space. This paper leverages these advances to build a system that understands influencers based on what they genuinely create, not merely how popular they appear.

II. RELATED WORK

Early formalization of influencer discovery includes the SoMin.ai framework [1], which leveraged behavioral analytics for automated profiling, and Gomes and Dias [2], who modeled influencer–merchant pairing as a stable matching problem. Neither approach incorporated multimodal content representation.

In recommender systems, Visual Bayesian Personalized Ranking (VBPR) [4] demonstrated that visual features significantly improve recommendation quality. Comprehensive surveys on Multimodal Recommender Systems confirm that jointly modeling text and visual signals provides richer semantic understanding [5]. Within the influencer domain, Kim et al. [3] proposed a Multimodal Post Attentive Profiling model combining captions and images for influencer categorization, the closest prior work to our approach.

A systematic literature review of 28 influencer marketing studies identifies the absence of unified systems integrating multimodal semantic matching with structured transactional workflows as the dominant research gap [11]. Our system directly addresses this gap.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Four-Layer Architecture

The platform follows a modular four-layer architecture as shown in Fig. 1. The **Data Layer** stores all influencer records in Supabase (PostgreSQL) and embedding datasets in Apache Parquet files on Google Drive (9.6M text + 1.6M image embeddings). The **AI Processing Layer** performs OpenCLIP (ViT-B/32) feature extraction and Multi-Head Attention Profiler inference to produce 256-dimensional influencer embeddings. The **Matching and Ranking Layer** retrieves top- K influencers using cosine similarity via pgvector’s IVFFlat

TABLE I
RAW OPENCLIP EMBEDDING CLUSTER QUALITY ($K=8$)

Metric	Text (512-d)	Image (512-d)
Silhouette Score	0.1600	0.1777
Cluster Purity	50.0%	74.3%
NMI	0.1429	0.4714
ARI	0.0719	0.3923

TABLE II
CLASSIFICATION PERFORMANCE — HELD-OUT VALIDATION SET

Metric	v1	v2	v3
Validation Accuracy	85.07%	85.49%	84.71%
Macro Avg F1	0.8373	0.8379	0.8296
Weighted Avg F1	0.8511	0.8553	0.8476
Pet F1	0.9050	0.9067	0.8841
Fashion F1	0.8810	0.8894	0.8832
Fitness F1	0.7447	0.7265	0.7377

42.2% (11,913 influencers); Pet is smallest at 2.1% (587). Training used 27,503 labeled influencers on an 80/20 stratified split (5,501 validation). Text embeddings were generated for 100% of 9,521,817 captions; image embeddings for ≈ 1.6 M posts (13% of dataset).

B. Pre-Training Embedding Analysis

K-Means clustering ($K=8$) on raw OpenCLIP embeddings revealed complementary modality strengths (Table I). Image embeddings show substantially stronger natural clustering (+230% relative NMI improvement). Categories where text clustering fails (pet, travel, interior) are well-structured in the image space, and vice versa—directly motivating the multimodal fusion design.

C. Neural Network Training Results

Three configurations were trained: v1 (1 block, 396K params, LAION text weights), v2 (4 blocks, 2.37M params, OpenAI weights for both modalities), and v3 (4 blocks, higher regularization, 20 epochs). Classification performance on 5,501 unseen validation influencers is shown in Table II. The v2 model achieves the best accuracy (85.49%) and F1 scores. Fitness is the most challenging category, primarily misclassified into fashion due to athleisure content overlap.

D. Retrieval Evaluation

Retrieval metrics were computed on the full 5,501-influencer validation set (Table III). All influencers were unseen during training. The v2 model achieves an 8.78 percentage point improvement in Precision@10 over v1 (72.06% $\rightarrow$ 80.84%). An MRR of 0.8612 means the most relevant influencer consistently appears at rank 1 or 2. Hit Rate@20 of 97.15% confirms that for 97 out of 100 brand queries at least one highly relevant influencer appears in the top 20 results.

Embedding space geometry reinforces these findings (Table IV). The v2 model achieves an inter/intra-class separation

TABLE III
RETRIEVAL PERFORMANCE AT $K=10$

Metric	v1	v2	v3
Precision@10	72.06%	80.84%	80.06%
NDCG@10	0.7198	0.8072	0.8003
MRR@10	0.7975	0.8612	0.8595
Hit Rate@10	93.72%	95.89%	95.73%
Hit Rate@20	95.70%	97.15%	96.95%

TABLE IV
EMBEDDING SPACE GEOMETRY

Metric	v1	v2	v3
Mean Intra-class Dist.	0.3198	0.1957	0.2147
Mean Inter-class Dist.	0.7227	0.7062	0.7021
Separation Ratio	2.26x	3.61x	3.27x

ratio of 3.61x vs. 2.26x for v1, with intra-class distance reduced from 0.3198 to 0.1957, confirming a well-structured 256-d representation space and explaining the jump in Precision@10.

E. End-to-End Pipeline Validation

The complete pipeline was validated on @zara (61.5M followers). The model correctly predicted FASHION with 97.7% confidence and all 10 nearest neighbours achieved cosine similarity ≥ 0.999 . Brand campaign simulation confirmed that image embeddings outperform text embeddings on visually distinctive queries (e.g., luxury skincare: 20% $\rightarrow$ 70% Precision@10), validating the multimodal fusion design.

VI. FUTURE WORK

Future work will extend the platform into a production-ready marketplace with: (1) **Full-dataset retraining** on all 9.5M posts once image embedding generation is complete; (2) **Layer 2 business-aware re-ranking** incorporating audience alignment (25%), budget compatibility (20%), engagement quality and fraud detection (20%), geographic fit (10%), influencer rating (10%), brand safety (8%), and content quality (7%) [8], [9]; (3) **Structured booking and milestone workflow**; (4) **Secure payment integration** with escrow-style release logic; (5) **WebSocket-based real-time in-app messaging**; (6) **Review and reputation system** feeding into Layer 2 re-ranking; and (7) **Cloud deployment** on Render/Railway and Vercel.

VII. CONCLUSION

This paper presents an AI-powered influencer-brand matching platform that replaces informal metric-driven selection with content-aware semantic matching. EDA of 9.5M posts from 33,935 influencers produced a clean 8-category dataset. The Multi-Head Attention Profiler achieves 85.49% classification accuracy and 80.84% Precision@10 on 5,501 unseen validation influencers, with an MRR of 0.8612 and a 3.61x embedding space separation ratio. End-to-end pipeline validation on @zara confirmed autonomous onboarding with

97.7% confidence. The Supabase database, FastAPI backend, and React.js frontend are fully operational. The system is technically sound and well-positioned for the full marketplace implementation described in the future work.

REFERENCES

- [1] A. Farseev, K. Lepikhin, H. Schwartz, E. K. Ang, and K. Powar, "SoMin.ai: Social Multimedia Influencer Discovery Marketplace," in *Proc. 26th ACM Int. Conf. Multimedia (MM '18)*, Seoul, 2018, pp. 1234–1236.
- [2] J. M. Gomes and L. A. V. Dias, "Recommending Influencers to Merchants Using Matching Game Algorithm," *arXiv preprint arXiv:2312.11498*, Dec. 2023.
- [3] S. Kim, J.-Y. Jiang, M. Nakada, J. Han, and W. Wang, "Multimodal Post Attentive Profiling for Influencer Marketing," in *Proc. The Web Conf. (WWW '20)*, Taipei, Apr. 2020, pp. 2878–2884.
- [4] R. He and J. McAuley, "VBPR: Visual Bayesian Personalized Ranking from Implicit Feedback," in *Proc. AAAI Conf. Artificial Intelligence*, 2016.
- [5] J. Xu et al., "A Survey on Multimodal Recommender Systems: Recent Advances and Future Directions," *arXiv preprint arXiv:2502.15711*, 2025.
- [6] C. J. Aarthy and K. Balaji, "The Evolution of Influencer Marketing: From Follower Counts to Authentic Brand Partnerships," *Power System Technology*, vol. 49, no. 1, pp. 1240–1254, 2025.
- [7] M. Beichert, A. Bayerl, J. Goldenberg, and A. Lanz, "Revenue Generation Through Influencer Marketing," *Journal of Marketing*, vol. 88, no. 4, pp. 40–63, 2024.
- [8] Nowadays Media, "Creator Selection Scorecard: A Data-Driven Influencer Vetting Framework," Mar. 2026. [Online]. Available: <https://nowadays.media/blog/creator-selection-scorecard-data-driven-framework/>
- [9] "Developing Influencer Selection Criteria for Brand-Influencer Collaboration," *Annals of Computer Science and Information Systems*, vol. 34, 2024.
- [10] P. Li, W. Zhan, L. Gao, S. Wang, and L. Yang, "Multimodal Recommendation System Based on Cross Self-Attention Fusion," *Systems*, vol. 13, no. 1, p. 57, 2025.
- [11] "A Systematic Literature Review on Influencer Marketing in the Digital Age," *Int. J. Multidisciplinary Research*, 2026.